PRAISING HARRIS

BLUE TEAM / DETECTION ENGINEERING / SOC

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2 Merged Tier-1 PRs SQLGlot & LiteLLM · **16 Upstream PRs** Anthropic MCP, LiteLLM, SQLFluff · **35.7% → 100%** ZeroTrace Sigma recall boost (0% FPR) · **Top 2%** TryHackMe, global

PROFILE

Undergraduate building detection engineering tooling, available for a 6-month full-time industrial co-op (Nov 14, 2026 – Summer 2026 with university NOC clearance and credit eligibility): behavioural baselining over Sysmon / EVTX, Sigma rules written from public threat reports, and coverage validation mapped to ATT&CK. Keeps LLMs out of the detection path: mappings come from a rulebook and narratives must be grounded in evidence.

EDUCATION

Indian Institute of Information Technology, Kottayam 2024 – 2028 (expected)
B.Tech Computer Science & Engineering (Cybersecurity) | CGPA 7.9
6-Month Co-op Availability: Nov 14, 2026 – Summer 2026 | University NOC Approved & Academic Credits Eligible

EXPERIENCE

VAPT Intern (Remote), Cyberops Jul 2026 – Oct 2026

- Completed company-assigned vulnerability assessment and penetration testing training covering Burp Suite, Nmap and the OWASP Top 10, practised on TryHackMe and HackTheBox machines.

PROJECTS

Upstream Security Hardening & Protocol Patches: MCP SDK, LiteLLM, SQLGlot github.com/Pyhroff
SQLGlot (Merged #8780) · LiteLLM (Merged #47091) · FastMCP (#6063) · Anthropic MCP SDK · CrewAI

- Shipped merged engine fix in BerriAI/litellm (PR #47091, merged), patched CVSS 7.5 multi-tenant authorization bypass (#47105), and committed fail-closed gateway tool resolution with machine-checked Lean 4 formal proof certificate (MCP_04.lean); hardened Anthropic's MCP Python SDK against RFC 6749 OAuth violations (PRs #3764, #3765, 175 tests).
- PrefectHQ/fastmcp (PR #6063) & SQLGlot: Enforced wall-clock timeouts and external callback caps mitigating host starvation in CodeMode (65/65 tests pass); shipped upstream AST node isolation in Toby Mao's SQLGlot (PR #8780, merged); authored 4 compiler grammar fixes in SQLFluff (#8899–#8902).

ZeroTrace: ATT&CK Telemetry Deobfuscator & AD Min-Cut Defense github.com/Pyhroff/zerotrace
Python · Sysmon EVTX · Edmonds-Karp Min-Cut · CLI Deobfuscation · 10/10 tests pass

- Behavioral Telemetry Deobfuscation: Built 5-layer CLI deobfuscator (Base64 -EncodedCommand, cmd.exe caret ^, backtick, quote fragmentation, [char] hex) normalizing Sysmon Event ID 8/10 traces; boosted naive Sigma rule recall from 35.7% to 100.0% at 0.0% FPR (F1 = 1.000) across 40 evasion samples.
- Active Directory Chokepoint Defense: Modeled BloodHound enterprise privilege graphs and applied Edmonds-Karp maximum-flow exact min-cut edge analysis; severed 100.0% of Tier-0 attack paths (180 → 0) by revoking only 7.15% (44/615) remediable ACL edges across 15 enterprise topologies in 6.09ms.

SOC PARALLAX: Behavioural Detection Platform github.com/Pyhroff/soc-parallax
Python · FastAPI · LangGraph · Neo4j · PostgreSQL · Next.js · Docker

- Built a detection pipeline over Sysmon / EVTX telemetry with per-entity (user, host, process) baselines, attributable risk scores where every point traces to a named signal, and ATT&CK mapping from a versioned YAML rulebook instead of LLM output.
- Correlated detections into incidents in a Neo4j graph and added a grounding guard that blocks LLM-written narratives from citing any IP or technique absent from the evidence; evaluated on the public EVTX-ATTACK-SAMPLES corpus.

PurpleLens: Detection Coverage Validator github.com/Pyhroff/PurpleLens
Python · pySigma · ATT&CK Navigator · pytest · GitHub Actions

- Wrote a Sigma rule evaluator on top of pySigma's parser and ran synthetic events for 10 ATT&CK techniques through vendored SigmaHQ rules and SOC PARALLAX's rules, exporting the result as an ATT&CK Navigator layer; 26 tests.
- Found a real granularity gap: rundll32 with no arguments (T1202) was caught by the Sigma rule but attributed to the broader T1218 by SOC PARALLAX.

Sigma Detection Rules: APT36 / SideCopy
SigmaHQ fork · Sigma · ATT&CK

- Authored three Sigma rules from public vendor threat reports: mshta fetching a remote HTA via an index.php endpoint, XenoRAT scheduled-task persistence, and CrimsonRAT / ElizaRAT registry persistence, tagged to ATT&CK techniques.

DarkDecoder: AI Threat Intelligence Platform github.com/Pyhroff/darkdecoder

Python · MITRE ATT&CK · MITRE ATLAS

- Maps threat-actor activity to ATT&CK and ATLAS techniques, extracts IOCs from unstructured threat reports and chains techniques to reconstruct kill chains, with LLM output grounded to prevent invented TTPs.

ModelHawk: ML Model Backdoor Scanner github.com/Pyhroff/ModelHawk

Python (stdlib only) · pytest · GitHub Actions

- Static scanner that detects pickle-deserialization code execution in PyTorch model files by reading the opcode stream (REDUCE / __reduce__ chains) without loading the model; 8 tests. Write-up published on Medium.

aisec-suite 0.3.0: Defensive AI-Security Scanner with SARIF CI Gate github.com/Pyhroff/aisec-suite

Python · Typer · AST taint analysis · SARIF · GitHub Actions · PyPI

- Published a unified scanner for MCP servers, agent memory files and RAG corpora: baseline mode fails CI only on new findings, reachability analysis cuts false positives (zero wrong refutations on 120 labelled findings), and the tool-poisoning detector reaches 48.1% held-out recall at 0.03% false positives on 3,974 real tool descriptions.
- Hand-checked results on about 90 public MCP-server repos and reported the false-positive rate (37.5% of HIGH permission_scope findings accurate in v0.6), then shipped patched versions.

SKILLS

Detection MITRE ATT&CK, Sigma, pySigma, EVTX / Sysmon, behavioural baselining, threat hunting, incident correlation (Neo4j), SARIF

Offensive knowledge Burp Suite, Nmap, OWASP Top 10, SQLi / LFI / XSS testing, recon and OSINT, Linux privilege escalation

Engineering Python, JavaScript, SQL, Bash; FastAPI, pytest, GitHub Actions, Docker, PostgreSQL

CERTIFICATIONS, WRITING & COMPETITIONS

TryHackMe top 2% globally, 177 rooms completed; Pre-Security and AI Security paths (Jun 2026). Fortinet: Introduction to the Threat Landscape (NSE 1).

Medium six security articles, including a static scanner for malicious ML models and agent-security write-ups. CTFs: participated in several competitions.

Hackathons PSBs Hackathon 2026 (PhantomGrid, behavioural-biometric authentication using Isolation Forest and DTW, with replay defence and a hash-chained audit log); USAII Global AI 2026.